Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 110.174.95.151

IS : 10553 (Part 5) - 1987

(Reaffirmed 2002)

(Reaffirmed 2012)

(Reaffirmed 2017)

(Reaffirmed 2022)

Indian Standard

REQUIREMENTS FOR CHLORINATION EQUIPMENT

PART 5 BLEACHING POWDER SOLUTION FEEDER DISPLACEMENT TYPE CHLORINATOR

UDC 628·162·841·06

© *Copyright* 1987

BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

Gr 2

December 1987

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

Indian Standard

REQUIREMENTS FOR CHLORINATION EQUIPMENT

PART 5 BLEACHING POWDER SOLUTION FEEDER DISPLACEMENT TYPE CHLORINATOR

Public Health Engineering Equipment Sectional Committee, BDC 40

Chairman	*Representing*
DR B. B. SUNDARESAN	University of Madras, Madras
Members	
ADVISER (PHE)	Ministry of Urban Development, New Delhi
ASSISTANT ADVISER (PHE) (*Alternate*)	
DR ALOKE DE	Jadavpur University, Calcutta
SHRI R. C. P. CHAUDHARY	Engineers India Limited, New Delhi
SHRI D. K. CHOUDHARY	Geo-Miller and Company, Calcutta
SHRI N. MANIVANNAN (*Alternate*)	
SHRI S. DAIVAMANI	Madras Metropolitan Water Supply and Sewerage Board, Madras
CHIEF ENGINEER (OPERATION AND MAINTENANCE) (*Alternate*)	
DEPUTY MUNICIPAL COMMISSIONER (SPECIAL ENGINEERING)	Municipal Corporation of Greater Bombay, Bombay
CHIEF ENGINEER (SEWERAGE) (*Alternate* I)	
HYDRAULIC ENGINEER (*Alternate* II)	
ENGINEER-IN-CHIEF	Delhi Water Supply and Sewage Disposal Undertaking, New Delhi
SHRI R. S. GAITONDE	Hydraulic and General Engineering Pvt Ltd, Bombay
SHRI M. RAVI SHANKAR (*Alternate*)	
SHRI K. K. GANDHI	Public Health Engineering Department, Government of Haryana, Chandigarh
SHRI M. N. SHARMA (*Alternate*)	
DR T. K. GEORGE	College of Engineering, Trivandrum
SHRI S. S. KALSI	Public Health Engineering Department, Government of Punjab, Patiala
SHRI R. A. KHANNA	Public Health Engineering Department, Government of Madhya Pradesh, Bhopal
SHRI G. S. RAGHVENDRA (*Alternate*)	

(*Continued on page* 2)

© *Copyright* 1987
BUREAU OF INDIAN STANDARDS
This publication is protected under the *Indian Copyright Act* (XIV of 1957) and reproduction in whole or in part by any means except with written permission of the publisher shall be deemed to be an infringement of copyright under the said Act.

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

(*Continued from page* 1)

Members	*Representing*
SHRI A. R. MIR	Public Health Engineering Department, Government of Jammu and Kashmir, Srinagar
SHRI G. M. KANTH (*Alternate*)	
SHRI U. N. MONDAL	Calcutta Metropolitan Development Authority, Calcutta
SHRI S. R. MUKHERJEE (*Alternate*)	
SHRI Y. N. NANJUNDIAH	Health and Family Welfare Department, Government of Gujarat, Gandhinagar
SHRI R. NATARAJAN	Hindustan Dorr-Oliver Limited, Bombay
SHRI SUBHASH VERMA (*Alternate*)	
PROF K. J. NATH	All India Institute of Hygiene and Public Health, Calcutta
SHRI D. GUIN (*Alternate*)	
SHRI A. N. RAO	Directorate General of Technical Development, New Delhi
SHRI K. P. DOHARE (*Alternate*)	
DR B. SUBBA RAO	Walchand College of Engineering, Sangli
SHRI K. PRABHAKAR RAO	Ministry of Defence (Engineer-in-Chief's Branch, Army Headquarters)
LT-COL M. P. THOMAS (*Alternate*)	
REPRESENTATIVE	Indian Institute of Technology, Kanpur
DR A. V. S. PRABHAKARA RAO (*Alternate*)	
REPRESENTATIVE	U. P. Jal Nigam, Lucknow
REPRESENTATIVE	Tamil Nadu Water Supply and Drainage Board, Madras
SHRI S. A. JAGDESAN (*Alternate*)	
SHRI A. SELVARAJ	Public Health Engineering Department, Government of Karnataka, Bangalore
SHRI KISHORE K. SANGHANI	Ahmedabad Municipal Corporation, Ahmadabad
SHRI A. K. SETH	National Environmental Engineering Research Institute, Nagpur
SHRI R. PARAMASIVAM (*Alternate*)	
SHRI S. A. SWAMY	Institution of Engineers (India), Calcutta
SHRI P. S. TENDOLKAR	Candy Filters (India) Limited, Bombay
SHRI C. L. SASTRI (*Alternate*)	
SHRI P. S. WADIA	Hindustan Construction Co Ltd, Bombay
SHRI C. E. S. RAO (*Alternate*)	
SHRI G. RAMAN, Director (Civ Engg)	Director General, BIS (*Ex-officio Member*)

Secretary

SHRI A. K. AVASTHY,
Deputy Director (Civ Engg), BIS

(*Continued on page* 8)

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

Indian Standard

REQUIREMENTS FOR CHLORINATION EQUIPMENT

PART 5 BLEACHING POWDER SOLUTION FEEDER DISPLACEMENT TYPE CHLORINATOR

0. FOREWORD

0.1 This Indian Standard (Part 5) was adopted by the Bureau of Indian Standards on 30 April 1987, after the draft finalized by the Public Health Engineering Equipment Sectional Committee had been approved by the Civil Engineering Division Council.

0.2 Bleaching powder solution type chlorinators are generally used in water treatment plants having capacities up to 1 MLD and in rural water supplies for disinfection of drinking water. This standard is intended to cover essential features of bleaching powder solution feeder displacement type chlorinator.

0.3 Bleaching powder is a variable mixture of calcium hydroxide, calcium chloride and calcium hypochlorite. When mixed with water, calcium hypochlorite dissociates into calcium hydroxide and hypochlorous acid (HOCl). Bleaching powder is made into a thin slurry with water and the supernatent which contains chlorine in solution is applied to water using a differential pressure dosing equipment.

0.4 For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS : 2-1960*. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

---

1. SCOPE

1.1 This standard (Part 5) lays down requirements for sizes, material and construction of bleaching powder solution feeder displacement type chlorinators.

*Rules for rounding off numerical values (*revised*).

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

2. CHLORINATOR CONSTRUCTION

2.1 The displacement type chlorinator consists of:

a) A fabricated mild steel pressure vessel with a removable flanged cover, drain and outlet valves and flexible plastic or rubber bag secured tight at its neck into which the bleaching powder solution is filled and from which it is fed to the pipeline carrying water to be chlorinated. The pressure vessel is designed for a working pressure of 0·175 N/mm^2 and is tested for 0·35 N/mm^2;

b) An orifice plate or venturi tube to produce a differential pressure in the pipeline. The differential pressure normally employed is 2 m of water head at maximum flow but can be varied, if necessary, to suit specific requirements;

c) Withdrawal injection fitting with isolating valve for insertion in the pipeline with necessary tubing to connect it with the pressure vessel. The fittings are to withstand the pressure within the doser;

d) A dose indicator and regulating valve attached to the pressure vessel; and

e) A solution preparing tank with necessary valves and pipings to enable the solution to flow by gravity into the pressure vessel.

A typical sketch of the assembly is shown in Fig. 1.

3. MATERIALS

3.1 Materials to be used in different components are given in Table 1.

4. SIZES

4.1 The size of the chlorinator is defined based on the capacity of the plastic or rubber bags. The chlorinators are of the following sizes:

a) 40 l b) 80 l c) 100 l d) 150 l e) 250 l

5. SELECTION OF DOSER SIZE

5.1 Capacity of doser is selected based on volume of bleaching powder solution required per day. Required volume of bleaching powder solution is calculated using the following formula:

$$v = \frac{0{\cdot}1\ Qx}{yz}$$

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151

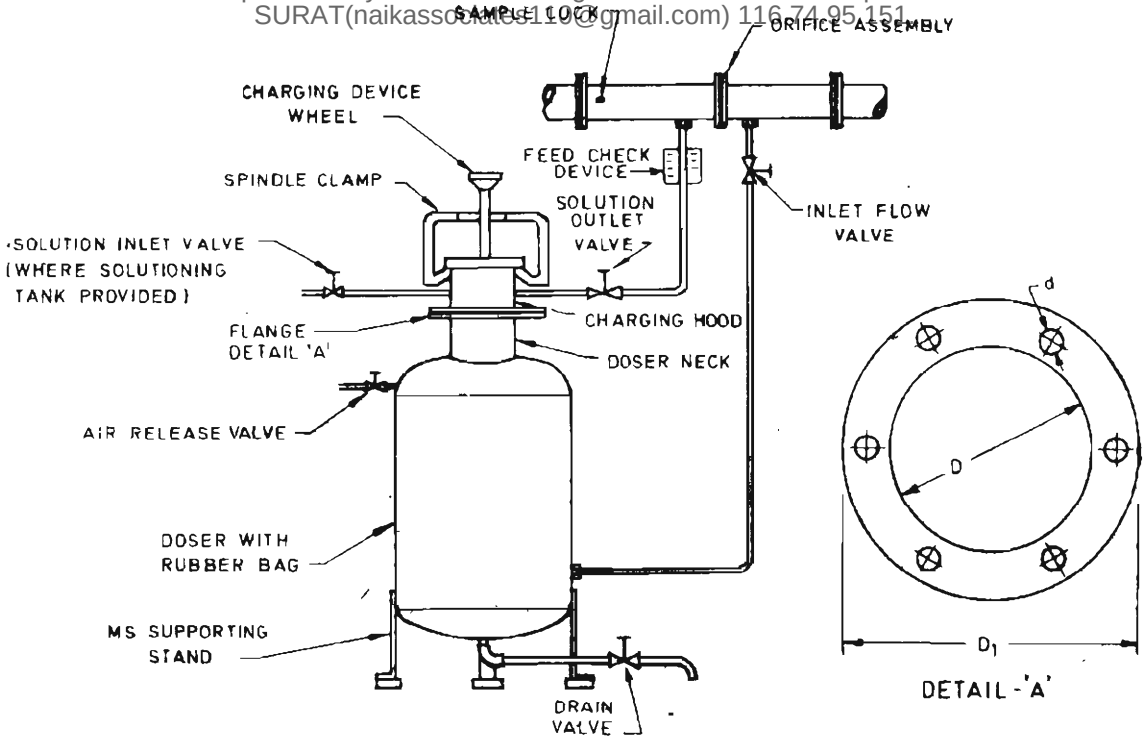

FIG. 1 TYPICAL SKETCH OF BLEACHING POWDER SOLUTION FEEDER DISPLACEMENT TYPE CHLORINATOR

NOTE 1 — For $D < 150$ mm, dia of nuts and bolts, d is 12 mm, and
For $D \geqslant 150$ mm, dia of nuts and bolts, d is 16 mm.

NOTE 2 — $D_1 = D + 2\,(4d)$.

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

where

v = volume of bleaching powder solution in litres,

Q = quantity of water to be disinfected per day (m^3),

x = chlorine dose to be applied (mg/l),

y = percent strength of bleaching powder solution (percent usually 3 to 5), and

z = available chlorine in bleaching powder expressed as a *fraction* (normally 0·25 to 0·3).

TABLE 1 MATERIALS OF DIFFERENT COMPONENTS OF BLEACHING POWDER SOLUTION FEEDER DISPLACEMENT TYPE CHLORINATOR

(*Clause* 3.1)

Sl No.	Component	Material	Reference to Indian Standard
(1)	(2)	(3)	(4)
i)	Doser shell	Mild steel with lining inside with rubber, PVC or epoxy	IS : 226-1975[1] IS : 4682 (Part 1)-1968[2] IS : 4682 (Part 4)-1969[3]
ii)	Bag	Natural or synthetic rubber/flexible plastic	—
iii)	Charging device operating wheel	Mild steel	IS : 226-1975[1]
iv)	Charging hood	Mild steel with lining inside with rubber, PVC or epoxy	IS : 226-1975[1] IS : 4682 (Part 1)-1968[2] IS : 4682 (Part 4)-1969[3]
v)	Air release valve	Plastic	IS : 9763-1981[4]
vi)	Feed check device/ rotameter:		
	a) Float	Polyethylene	IS : 9762-1981[5]
	b) Tube	Glass/PVC	IS : 7374-1974[6] IS : 3076-1984[7] IS : 4984-1987[8] IS : 4985-1987[9]
vii)	Inlet flow operating valve	Plastic	IS : 9763-1981[4]
viii)	Inlet flow regulating valve	Plastic	IS : 9763-1981[4]
ix)	Drain valve	Plastic	IS : 9763-1981[4]
x)	Charging device clamp	Mild steel	IS : 226-1975[1]
xi)	Air release vent pipe	PVC	IS : 3076-1984[7] IS : 4984-1987[8] IS : 4985-1987[9]
xii)	Spindle for doser	Mild steel	IS : 226-1975[1]

(*Continued*)

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

TABLE 1 MATERIALS OF DIFFERENT COMPONENTS OF BLEACHING POWDER SOLUTION FEEDER DISPLACEMENT TYPE CHLORINATOR — *Contd*

Sl No.	Component	Material	Reference to Indian Standard
(1)	(2)	(3)	(4)
xiii)	Drain pipe or bend	PVC	IS : 4984-1987[8] IS : 4985-1987[9] IS : 10124 (Part 8)-1982[10]
xiv)	Socket for rotameter, drain, air release vent	PVC	IS : 7834 (Part 1)-1975[11]
xv)	Orifice assembly (where provided)	Corrosion resistance steel	Grade 6 of IS : 3444-1978[12]

[1]Specification for structural steel (standard quality) (*fifth revision*).

[2]Code of practice for lining of vessels and equipment for chemical processes: Part 1 Rubber lining.

[3]Code of practice for lining of vessels and equipment for chemical processes: Part 4 Plasticized PVC.

[4]Specification for plastic bib taps and stop valves (rising spindle) for cold water services.

[5]Specification for polyethylene floats for ball valves.

[6]Specification for glass rods and tubing.

[7]Specification for low density polyethylene pipes for potable water supplies (*second revision*).

[8]Specification for high density polyethylene pipes for potable water supplies, sewage and industrial effluents (*third revision*).

[9]Specification for unplasticized PVC pipes for potable water supplies (*third revision*).

[10]Specification for fabricated PVC fittings for potable water supplies: Part 8 Specific requirements for 90° bend.

[11]Specification for injection moulded PVC socket fittings with solvent cement joints for water supplies: Part 1 General requirements.

[12]Specification for corrosion resistant alloy steel and nickel based castings for general applications (*first revision*).

Free Standard provided by BIS via BSB Edge Private Limited to Krupaben Brahmbhatt - SURAT(naikassociates110@gmail.com) 116.74.95.151.

(*Continued from page* 2)

Water Treatment Equipment Subcommittee, BDC 40 : 1

Convener	*Representing*
SHRI JAI NARAIN	Indian Water Works Association, New Delhi
Members	
ASSISTANT ADVISER (PHE)	Ministry of Urban Development, New Delhi
DR A. G. BHOLE	Visvesvarya Regional College of Engineering, Nagpur
DR S. M. DHABADGAONKAR (*Alternate*)	
SHRI S. N. CHAKRABARTY	Engineers India Limited, New Delhi
SHRI D. K. CHOUDHURY	Geo-Miller and Company Pvt Ltd, Calcutta
SHRI A. K. MAZUMDAR (*Alternate*)	
SHRI D. D. DESAI	Pennwalt India Limited, Bombay
SHRI V. P. THANGAMUTHU (*Alternate*)	
SHRI R. S. GAITONDE	Hydraulic and General Engineer Pvt Ltd, Bombay
SHRI V. M. SHIRODKAR (*Alternate*)	
SHRI N. W. MIRACHANDANI	Shah Technical Consultants Pvt Ltd, Bombay
SHRI R. NATARAJAN	Hindustan Dorr-Oliver Limited, Bombay
SHRI SUBHASH VERMA (*Alternate*)	
PROF K. J. NATH	All India Institute of Hygiene and Public Health, Calcutta
SHRI D. GUIN (*Alternate*)	
SHRI R. PARMASIVAM	National Environmental Engineering Research Institute, Nagpur
SHRI N. G. SWARNAKAR (*Alternate*)	
SHRI N. RAMACHANDRAN	Ion Exchange (India) Ltd, Bombay
SHRI A. M. JOSHI (*Alternate*)	
SHRI C. E. S. RAO	Hindustan Construction Company Ltd, Bombay
REPRESENTATIVE	Municipal Corporation of Greater Bombay, Bombay
SHRI C. L. SASTRI	Candy Filters (India) Ltd, Bombay
SHRI V. SATYAMURTHI	EIMCO-KCP Ltd, Madras
SHRI T. SUBRAHMANAYAM (*Alternate*)	
SHRI S. S. SRIVASTAVA	U. P. Jalnigam, Lucknow
SHRI S. A. SWAMY	Delhi Water Supply and Sewage Disposals Undertaking, New Delhi
SHRI S. S. RAMRAKHYANI (*Alternate*)	